

The empirical recurrence for OEIS A252029, recovered from its tail

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Abstract

OEIS A252029 records “Empirical recurrence of order 83” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 164$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 89$. Its last coefficient is $c_{83} = -240 \neq 0$, so no further tail satisfies anything shorter; any order-83 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A252029 is “Number of $(n+2) \times (5+2)$ 0..2 arrays with every 3×3 subblock row, column, diagonal and antidiagonal sum not equal to 0 2 or 4.”. It has offset 1 and begins

1805, 5285, 24664, 126920, 672903, 3560831, 18791972, 99214423, ...

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 10:42:58, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 164$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 164$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 83: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 83.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 328$ exact terms from $n = 6$ on, it returns order 83 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	7	c_{22}	63708794	c_{43}	−8151028904	c_{64}	53483823
c_2	−17	c_{23}	169098800	c_{44}	2736149095	c_{65}	52415918
c_3	71	c_{24}	191481067	c_{45}	11475543676	c_{66}	21820244
c_4	−98	c_{25}	−77774659	c_{46}	11547650516	c_{67}	−7297805
c_5	−191	c_{26}	−542180456	c_{47}	6942112771	c_{68}	−17336878
c_6	−448	c_{27}	−814135251	c_{48}	251914467	c_{69}	−12078037
c_7	−2392	c_{28}	−378979986	c_{49}	−2943868986	c_{70}	−3855329
c_8	7041	c_{29}	1009444374	c_{50}	−2817477811	c_{71}	1047419
c_9	14848	c_{30}	2397010079	c_{51}	−735334309	c_{72}	1677312
c_{10}	23574	c_{31}	2221947105	c_{52}	621245666	c_{73}	902495
c_{11}	5165	c_{32}	−464217734	c_{53}	472194079	c_{74}	99160
c_{12}	−232577	c_{33}	−4380977529	c_{54}	−536038293	c_{75}	−142721
c_{13}	−384299	c_{34}	−6644004318	c_{55}	−1234192314	c_{76}	−142767
c_{14}	−297122	c_{35}	−3505103407	c_{56}	−1077206972	c_{77}	−68564
c_{15}	958226	c_{36}	3459260612	c_{57}	−426772065	c_{78}	−20424
c_{16}	4188774	c_{37}	11046967784	c_{58}	174189222	c_{79}	−2966
c_{17}	4286298	c_{38}	12282358279	c_{59}	335723499	c_{80}	588
c_{18}	−1877873	c_{39}	5544572986	c_{60}	219954295	c_{81}	−852
c_{19}	−18755348	c_{40}	−6321484793	c_{61}	35563297	c_{82}	−552
c_{20}	−39018646	c_{41}	−15478547063	c_{62}	−19770290	c_{83}	−240
c_{21}	−17602819	c_{42}	−16680594647	c_{63}	9787327		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 89$, and it is the recurrence OEIS A252029 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{83} - c_1x^{83-1} - \dots - c_{83}$. Its constant term is $-c_{83} = 240$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 83 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 83, so $\deg q = 83$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 83, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 83, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -240 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-83 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A252029.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.